

LESSON 11.6

REPRESENTING PROPORTIONAL RELATIONSHIPS WITH EQUATIONS



LESSON INTRODUCTION

MA.7.AR.4.2 Determine the constant of proportionality within a mathematical or real-world context given a table, graph, or written description of a proportional relationship.

MA.7.AR.4.4 Given any representation of a proportional relationship, translate the representation to a written description, table, or equation.

LEARNING TARGETS

- I can determine the constant of proportionality of a proportional relationship.
- I can translate a given representation of a proportional relationship between two quantities to an equation.

BENCHMARK COHERENCE

BUILDING ON

MA.6.AR.3.2 Given a real-world context, determine a rate for a ratio of quantities with different units. Calculate and interpret the corresponding unit rate.

MA.6.AR.3.5 Solve mathematical and real-world problems involving ratios, rates, and unit rates, including comparisons, mixtures, ratios of lengths, and conversions within the same measurement system.

WORKING TOWARD

MA.8.AR.3.2 Given a table, graph, or written description of a linear relationship, determine the slope.

MA.8.AR.3.3 Given a table, graph, or written description of a linear relationship, write an equation in slope-intercept form.

MA.8.AR.3.5 Given a real-world context, determine and interpret the slope and y-intercept of a two-variable linear equation from a written description, a table, a graph, or an equation in slope-intercept form.

ESSENTIAL QUESTIONS

- How can you write an equation to represent a proportional relationship?
- How can you translate a graph, table, or written description of a proportional relationship to an equation?

LESSON NARRATIVE

Students will be formally introduced to writing equations to represent proportional relationships seen in tables, graphs, and written descriptions. Students begin by interacting with this content in a student-guided 11.6.2 Exploration before synthesizing the takeaways through 11.6.4 Guided Instruction. The primary learning outcome of this lesson is for students to use the formula $y=kx$ and its derivatives to represent proportional relationships seen in tables, graphs, and written descriptions.

COMPONENT SUMMARY

11.6.1 WARM-UP (~5 MIN)

Use percentages to solve a problem

11.6.3 GUIDED INSTRUCTION (15 MIN)

Write equations to represent proportional relationships

11.6.5 YOUR TURN! (5 MIN)

Independent practice writing equations to represent proportional relationships

11.6.2 EXPLORATION (10-15 MIN)

Explore proportional relationships and equations

11.6.4 GUIDED PRACTICE (5 MIN)

Practice writing equations to represent proportional relationships

11.6.6 WRAP-UP (~5 MIN)

Assess writing equations to represent proportional relationships

RESOURCES

GLOSSARY

Key Terms: N/A

Additional Terms:

- Equation
- Proportional relationship

MATERIALS

- N/A

ADDITIONAL PREPARATION

- N/A

TEACHER TIPS

COMMON MISCONCEPTIONS

- Students may struggle to connect concrete information in a graph, table, or written description to an abstract equation.
- Students may struggle to understand that $y=kx$ is a formula that can be used to represent any proportional relationship.

THINGS TO CONSIDER

- In the 11.6.2 Exploration, encourage students to consider how each part of a problem relates to the equation they create. Model this with students in the 11.6.3 Guided Instruction.
- Make $y=kx$ more concrete by substituting values for the different variables to relate the different representations.

LITERACY AND LANGUAGE ROUTINES

Take Turns

Consider implementing a Take Turns routine for the 11.6.4 Guided Practice. This activity provides the opportunity for students to “take turns” using pairs of values in a table, graph, or written description to determine a constant of proportionality and an equation to represent a proportional relationship. Using a Take Turns routine will give students the opportunity to explain and articulate their process for solving a problem.

MATHEMATICAL THINKING AND REASONING (MTR) STANDARD

MA.K12.MTR.3.1 Mathematical Fluency

Students will be working to write an equation to represent a proportional relationship. Clarifications within this standard emphasize using appropriate methods to solve problems and completing tasks with confidence. Utilize the progression of the lesson to allow students to practice efficient and generalizable mathematical methods.

DIFFERENTIATE INSTRUCTION

Encourage conversations and discussions about problem-solving strategies to support students who benefit from auditory learning and to support students learning English as a second language. Support students as they make connections between graphs or tables and abstract equations. Provide visual entry points through encouraging learners to notice the relationships between the variables and their graphs and tables.

11.6.1 WARM-UP: FIGHTING FOR EQUAL PAY

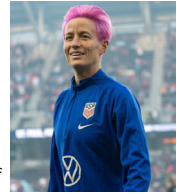
TEACHER MOVES

If time allows, engage students in a discussion of the context and their thoughts on the issue.



11.6.1 Warm-Up: Fighting for Equal Pay

Megan Rapinoe is an American professional soccer player and captain of the OL Reign, a team in the National Women's Soccer League (NWSL), as well as co-captain of the U.S. women's national team. She has won many awards and championships, including the 2019 FIFA Women's World Cup. In March of 2019, she and a group of 27 of her teammates filed a lawsuit against the U.S. Soccer Federation over gender discrimination as part of the team's fight for equal pay. The U.S. women's national team has been largely more successful and popular than the U.S. men's national team. Yet, as of 2019, the top women's players earned only 38% of the pay their top male counterparts earned.



1. In 2019, players from the men's national team earned an average of \$13,166 per game. How much did the average women's national team player earn per game at the time?

\$5,003.08 per game



11.6.2 Exploration: Proportional Relationships and Equations

Work with your partner to complete the following.

1. A recipe says that 2 cups of dry rice will produce 6 servings.

- a. How many servings will 1 cup of dry rice produce?
1 cup of dry rice will produce 3 servings.
- b. Complete the table.

Cups of Dry Rice, x	Number of Servings, y
1	3
2	6
3	9
12	36
43	129

- c. Discuss with your partner how you determined the number of servings 43 cups of dry rice would produce.
Answers will vary. If 1 cup produces 3 servings, you could multiply the cups of dry rice by 3 to get 129 servings.
 - d. How many servings does x cups of dry rice produce?
It produces $3x$ servings.
 - e. Write an equation that could be used to find the number of servings, y , that any number of cups of dry rice, x , could produce.
 $y = 3x$
2. On their party-planning menu, a catering company says that 6 spring rolls are considered 3 servings.
 - a. How many servings will 1 spring roll produce?
1 spring roll is $\frac{1}{2}$ a serving.

- b. Complete the table.

Number of Spring Rolls, x	Number of Servings, y
1	0.5
6	3
10	5
16	8
25	12.5

- c. With your partner, discuss what was similar and what was different about completing this table when compared with the table in Problem 1b. Summarize your discussion below.

How was completing the tables similar?	How was completing the tables different?
<i>Answers will vary. One way completing these tables was similar was once I found the constant of proportionality, all I had to do was find the missing y values by multiplying the x value by the constant of proportionality.</i>	<i>Answers will vary. One way completing these tables was different was in problem 1b the y values were larger than the x values, whereas in 2b the y values were smaller than the x values.</i>

- d. How many servings would x spring rolls be?
 $0.5x$ or $\frac{1}{2}x$
- e. Write an equation that could be used to find the number of servings, y , that any number of spring rolls, x , could produce.
 $y = 0.5x$ or $y = \frac{1}{2}x$

11.6.2 EXPLORATION: PROPORTIONAL RELATIONSHIPS AND EQUATIONS

ENRICHMENT

Ask guiding questions, if necessary, as students discuss their responses to questions 1c-1e.

- How did you determine the relationship of cups of rice to number of servings? Why is this helpful to know?
- If you know the number of cups of rice, how could you determine the number of servings? If you know the number of servings, how would you figure out the cups of rice?
- What is the constant of proportionality of the relationship?

DIFFERENTIATE INSTRUCTION

Consider allowing students to use different colors to connect the values in the equation with the values in the table. Have students use the same color to highlight or write the constant of proportionality in the equation and in the problem.

TEACHER MOVES

Encourage students to consider how each part of question 2 relates to the equation in part e. Ask students what x and y represent in the relationship. Guide students toward understanding that y is dependent on x . Also guide students toward connecting that the 0.5 used in the equation is the constant of proportionality in the relationship.

11.6.3 GUIDED INSTRUCTION: WRITING EQUATIONS TO REPRESENT PROPORTIONAL RELATIONSHIPS

DIFFERENTIATE INSTRUCTION

Use an “everything together” strategy to help students see how the graph, table, and equation connect. Encourage students to match each pair of values in the table with a point on the graph. Then have students determine the constant of proportionality using the table and graph and write it beside them. Lastly, guide students as they use the constant of proportionality to write the equation.

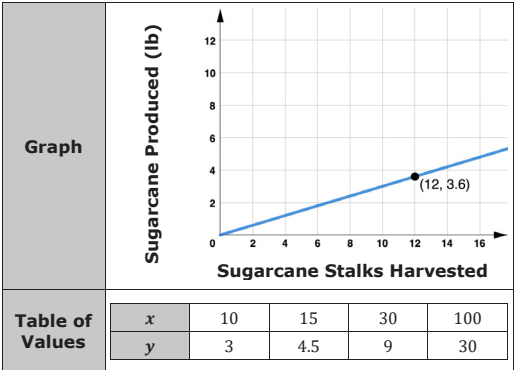
Encourage students to verbally explain what they are doing and to touch each point and pair of values to show their connection.

11.6.3 Guided Instruction: Writing Equations to Represent Proportional Relationships

In each situation from the Exploration, an equation can be written to represent the relationship.

When y is proportional to x , x can be multiplied by the same number to get y . The number multiplied by x is the constant of proportionality, k .
The equation representing this proportional relationship is $y = kx$.

1. More than half of the sugarcane produced in the United States comes from farms in Florida. Sugarcane is grown in stalks. The graph and table of values show the relationship between the amount of sugar produced in pounds (lb), y , and the number of stalks harvested, x .



- a. Complete the steps to determine the constant of proportionality, $\frac{y}{x}$, using the point on the graph.

$$\frac{y}{x} = \frac{3.6}{12} = 0.3$$

- b. Select any ordered pair from the table of values and use it to find the constant of proportionality, $\frac{y}{x}$.

$$\frac{y}{x} = \frac{3}{10} = 0.3$$

- c. Complete an equation that represents this relationship.

$$y = 0.3x$$

- d. Complete the written description of this proportional relationship.

The average ☐ sugarcane stalk ☐ pound of sugar produces 0.3

- ☐ sugarcane stalks.
☐ pounds of sugar.

When given a table of values or a graph, an equation representing the proportional relationship can be written by finding the constant of proportionality.

- e. Consider the equation $y = \frac{y}{x} \cdot x$. Work with your partner to explain why both sides of this equation are equivalent.

When $\frac{y}{x}$ is multiplied by x , the result is y because

$$\frac{y}{x} \cdot x = y$$

- f. An equation for a proportional relationship can also be written in terms of x . Complete the equation using a ratio that would make both sides of the equation equivalent.

$$x = \frac{x}{y} \cdot y$$

11.6.3 GUIDED INSTRUCTION: WRITING EQUATIONS TO REPRESENT PROPORTIONAL RELATIONSHIPS (CONTINUED)

TEACHER MOVES

When modeling how to write the equation, consider assigning students different ordered pairs for question 1b and then discussing/sharing as a whole class to reinforce that the constant of proportionality will be the same regardless of which point is used.

ENRICHMENT

Help students make connections between the visual representations and the constant of proportionality through questioning.

- How does the constant of proportionality relate to the graph?
- What is the y -value for an x -value of 6? How much sugarcane is produced from 10 sugarcane stalks?
- How can you use the constant of proportionality to find the y -value for any given x -value without looking at the graph?

TEACHER MOVES

To help students understand the relation between an equation and the proportional relationship it represents, guide students to substitute an x -value from a graph, table, or written description and then multiply it by the constant of proportionality.

11.6.3 GUIDED INSTRUCTION: WRITING EQUATIONS TO REPRESENT PROPORTIONAL RELATIONSHIPS (CONTINUED)

TEACHER MOVES

Two equations can be written to represent this relationship. However, modeling with the constant of proportionality determined in question 2a offers an opportunity to model how to relate the variables in the equation such that both expressions are equivalent. For example, if $\frac{s}{c} = 32.5$, then multiplying this ratio by c will result in s .

QUESTIONING

If time allows, ask students to work with a partner to determine the other equation that could be used to model the same relationship.

- g. Determine the constant of proportionality, $\frac{x}{y}$, for the relationship between the amount of sugar produced in pounds (lb), y , and the number of stalks harvested, x .

$$k = \frac{10}{3}$$

- h. Write another equation to represent the same relationship in terms of x .

$$x = \frac{10}{3}y$$



While two equations can be written to represent any proportional relationship, equations are most commonly written $y = kx$, where x is the independent variable, y is the dependent variable, and k is the constant of proportionality, $\frac{y}{x}$.

Any two variables can be used in the equation of a proportional relationship.

2. In 2018, there were 481,000 tons of sugar produced, s , from the 14,800 acres of sugarcane harvested, c , in Martin County, FL.

- a. Determine the constant of proportionality, $\frac{s}{c}$, from this written description.

$$\frac{s}{c} = \frac{481,000}{14,800} = 32.5$$

- b. Write an equation to represent this relationship.

$$s = 32.5c$$

11.6.4 GUIDED PRACTICE: WRITING EQUATIONS TO REPRESENT PROPORTIONAL RELATIONSHIPS

DIFFERENTIATE INSTRUCTION

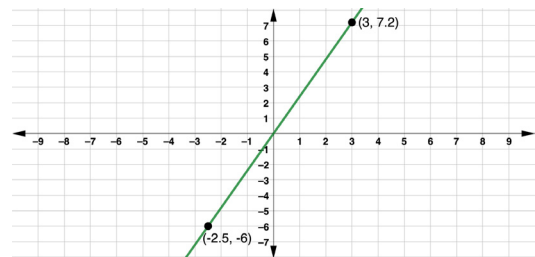
If students struggle to determine the equation, emphasize the relationship between the points on the graph and the equation. Students may also benefit from creating a table and a written description to further the connections.

They can verify their equation is accurate by substituting the coordinate pairs from the graph to verify.



11.6.4 Guided Practice: Writing Equations to Represent Proportional Relationships

1. The graph of a proportional relationship is shown.



Write an equation to represent the relationship.

$$y = 2.4x$$

2. The aluminum from recycled cans can be used to make other products. The aluminum from 32 cans that are recycled in Tampa, FL has a value of \$0.40. If a is the number of aluminum cans recycled and v is their value, in dollars, complete each equation to represent this relationship.

$$a = \boxed{80}v$$

$$v = \boxed{0.0125}a$$

**11.6.5 Your Turn!**

- A bakery uses 8 tablespoons of honey for every 10 cups of flour to make bread dough. If f is the number of cups of flour used and h is the number of tablespoons of honey, complete each equation to represent this relationship.

$$f = \boxed{\frac{5}{4}} h$$

$$h = \boxed{\frac{4}{5}} f$$

- The table shows the amount of red paint and blue paint needed to make a certain shade of purple paint called Venusian Sunset.

Red Paint (parts)	Blue Paint (parts)
3	12
7	28
1.5	6

If r is the amount of red paint used and b is the amount of blue paint used, complete each equation to represent this relationship.

$$b = \boxed{4} r$$

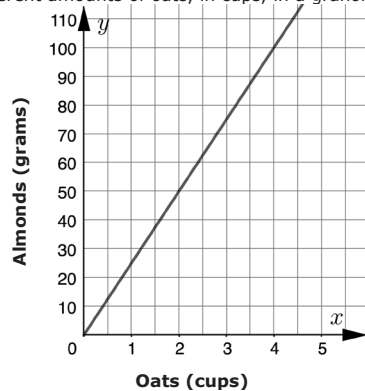
$$r = \boxed{\frac{1}{4}} b$$

11.6.5 YOUR TURN!**DIFFERENTIATE INSTRUCTION**

Consider allowing students to complete 11.6.4 Guided Practice and 11.6.5 Your Turn! with a partner or in small groups. Partners can utilize a Take Turns strategy, whereas small groups may benefit from using a jigsaw approach with each student solving one problem and then explaining how they solved it to the other group members.

**11.6.6 Wrap-Up: Ticket Out the Door**

- The graph shows the amounts of almonds, in grams, for different amounts of oats, in cups, in a granola mix.



Write an equation that represents the relationship between the number of cups of oats, x , and the number of grams of almonds, y , in the granola mix.

$$y = \boxed{25} x$$

11.6.6 WRAP-UP: TICKET OUT THE DOOR**DIFFERENTIATE INSTRUCTION**

The information provided from the students here, coupled with what was observed during instruction, can provide information for differentiating instruction in later lessons.